Balki Robocup — Test & Task Summary

Competition format, scoring, and category tasks for all teams.

How the event works

- **Kits are provided:** Each team receives a sealed kit with every component, tool, and program needed for its experiment. Teams don't bring or buy any parts.
- **One hour per team:** Each team gets 1 hour in total, covering both assembly and the task.
- **Judges watch the whole hour:** Judges record the assembly time and log every mistake as it happens.
- **Program:** The code is supplied with the kit. Teams may only change the calibration values marked in the code, such as sensor thresholds.

The 1-hour slot

1. **Kit check (first 5 minutes):** Open the kit and check every part against the parts list. Anything missing or damaged that is reported now is replaced with no deduction.
2. **Assembly:** Build and wire the experiment.
3. **Assembly check:** When the build is done, the team calls a judge. That clock time is recorded as the assembly time, and the judge inspects the wiring before the power is switched on.
4. **Calibration and attempts:** Teams calibrate and then make up to 2 task attempts. The better attempt counts. Each attempt has a maximum time, listed under each category.
5. **Hard stop at 60 minutes:** Anything not completed by then scores 0.

Scoring (100 points)

Assembly efficiency (40 points), the same for every category:

- **Assembly time (15 points):** 15 points for 25 minutes or less, 10 for 26–35 minutes, 5 for 36–45 minutes, and 2 for over 45 minutes.
- **First-time-right build (10 points):** 10 points if the build passes the judge's check with no corrections, 6 after one correction, 3 after two, and 0 for three or more.
- **Build quality (10 points):** Neat wiring, secure mounting, and no loose parts, judged at the assembly check.
- **Teamwork and organisation (5 points):** Parts laid out, work shared between members, and a tidy workspace.
- **Assembly mistakes** (deducted from the assembly score, which cannot go below 0):
  - –2 each time the team asks a judge for help with the build.
  - –3 for switching on the power before the judge's check.
  - –5 for a component damaged during assembly (one replacement is given).

Task performance (60 points), specific to each category: Three tasks of 20 points each. Mistakes during a task are deducted as listed under that task, and a task score cannot go below 0.

Tie-breakers

1. Fewer total mistakes (assembly and task combined).
2. Faster assembly time.
3. The category tie-breaker listed under each category.

Scorecard

Team	Assembly time (15)	First-time-right (10)	Build quality (10)	Teamwork (5)	Task 1 (20)	Task 2 (20)	Task 3 (20)	Total mistakes	Total (100)
Team 01									/ 100
Team 02									/ 100
Team 03									/ 100

1. Line-Follower & Obstacle Bot

Arena: A white sheet track made with 1.5–2" black tape, in three zones:

- **Zone A:** Curves, 90° turns, and a loop.
- **Zone B:** A black box blocking the line.
- **Zone C:** A double-line finish with a stop wall 15 cm behind it.

Task performance (60 points):

- **Task 1 – Line tracking (20 points):**
  - 12 points for finishing the track.
  - 8 points for smooth driving without leaving the line.
  - Mistakes: –3 per manual touch, and –5 if the line is lost for more than 5 seconds.
- **Task 2 – Obstacle avoidance (20 points):**
  - 8 points for detecting the box at 10–15 cm without hitting it.
  - 7 points for a smooth detour around it.
  - 5 points for finding the line again.
  - Mistakes: –10 for a crash, and –5 if the line isn't found again.
- **Task 3 – Speed and stop (20 points):**
  - A 1.5 m sprint scores 10 points for 5 seconds or less, 7 for 5.1–8 seconds, and 4 for over 8 seconds.
  - The stop is measured past the finish line: 10 points for 0–5 cm, 6 for 5–10 cm, and 0 for touching the wall.

Category rules: Each attempt can last up to 3 minutes.

Category tie-breaker: Faster Task 3 sprint time.

2. RC Bluetooth Smart Car

Arena: A 3 × 2 m floor area with three zones:

- **Slalom:** 5 cones set 30 cm apart.
- **Covered garage:** For blind reverse parking.
- **Drag strip:** 3 m long, with rails.

Task performance (60 points):

- **Task 1 – Slalom (20 points):**
  - 12 points for completing the course.
  - 8 points for smooth control.
  - Mistakes: –3 per cone hit, and –5 for going off track.
- **Task 2 – Reverse parking (20 points):**
  - 10 points for docking.
  - 5 points for reversing in.
  - 5 points for centring.
  - Mistakes: –4 for hitting a wall, and –2 for taking more than 3 tries to align.
- **Task 3 – Drag and brake (20 points):**

  - The sprint scores 10 points for 4 seconds or less, 7 for 4.1–6 seconds, and 4 for over 6 seconds.
  - Stopping inside the 20 cm zone scores 10 points; overshooting scores 0.

Category rules:

- Each attempt can last up to 3 minutes.
- Teams drive using the control app named in the kit instructions.
- If Bluetooth drops, the team has 30 seconds to re-pair while the timer keeps running. Otherwise the attempt ends with the points earned so far.

Category tie-breaker: Faster Task 3 sprint time.

3. ESP32 Wi-Fi Surveillance Bot

Arena:

- The driver sits behind a curtain and drives only from the live video feed.
- **Zone A:** A 3 × 2 m maze with 30 cm walls.
- **Zone B:** 4 hidden targets (QR codes, numbers, or symbols) placed 10–15 cm high inside the maze.
- **Zone C:** A 3 m corridor behind a barrier, to test the signal.

Task performance (60 points):

- **Task 1 – FPV maze (20 points):**
  - 12 points for reaching the exit.
  - 8 points for a smooth video stream.
  - Mistakes: –3 per wall hit, and –5 for looking over the curtain.
- **Task 2 – Target recon (20 points):**
  - 16 points for identifying targets (4 each).
  - 4 points for a clear, focused image.
  - Mistakes: –3 per misread target.
- **Task 3 – Range and latency (20 points):**
  - 10 points for a response time of 200 ms or less.
  - 10 points for keeping the video running behind the barrier.
  - Mistakes: –5 for a freeze longer than 3 seconds, and –10 for a server crash.

Category rules:

- Each attempt can last up to 4 minutes.
- The web dashboard is supplied with the kit code.
- One emergency reboot is allowed per attempt (a 60-second window, with the timer running).

Category tie-breaker: Higher Task 2 score.

4. Smart Plant / Agri Mini-Lab

Test bench: Three soil cups plus a 500 mL reservoir and a measuring cylinder:

- Dry soil (below 15% moisture).
- Moist soil (40–60%).
- Waterlogged soil (80–100%).

Task performance (60 points):

- **Task 1 – Calibration and display (20 points):**
  - 8 points for accurate moisture percentages.
  - 8 points for an LCD showing moisture %, pump status, and live updates.
  - 4 points for stable, noise-free readings.
  - Mistakes: –5 for a garbled or frozen display.
- **Task 2 – Automatic watering (20 points):**
  - 8 points for the pump turning on within 2 seconds in dry soil.

  - 8 points for a clean cut-off in moist and wet soil.
  - 4 points for no relay chatter near the threshold.
  - Mistakes: –10 if the pump fails to switch either way, and –5 if a reset is needed between cups.
- **Task 3 – Water precision (20 points):**
  - 12 points for delivering 30–50 mL and stopping.
  - 8 points for no spills or leaks.
  - Mistakes: –5 for overflow, and –3 for leaking tubing.

Category rules:

- Each attempt can last up to 3 minutes.
- If water spills on the electronics, the attempt pauses and the team gets 60 seconds to dry them and resume with no deduction.

Category tie-breaker: Higher Task 2 score.

5. 2-DOF Robotic Gripper Arm

Test station:

- A 60 × 40 cm table.
- **Pick-up zone:** A paper cup (10 g), a ping-pong ball (2.7 g), and a block (45 g).
- **Drop-off:** 2 rings, at a 90° turn from the pick-up zone.
- **Stacking platform:** 5 cm high.

Task performance (60 points):

- **Task 1 – Pick and place (20 points):**
  - 6 points for the cup, 7 for the ball, and 7 for the block.
  - Mistakes: –3 for a drop, and –2 for landing outside a ring.
- **Task 2 – Stacking (20 points):**
  - 10 points for placing a block on the platform.
  - 10 points for stacking 2 blocks.
  - Mistakes: –5 for a collapse, and –3 for touching an object by hand.
- **Task 3 – Speed and smoothness (20 points):**
  - A 3-object circuit scores 12 points for 60 seconds or less, 8 for 61–120 seconds, and 4 for over 120 seconds.
  - Up to 8 points for smooth, jitter-free movement.

Category rules:

- Each attempt can last up to 3 minutes.
- The arm base must be clamped or weighted; teams cannot hold it down by hand.

Category tie-breaker: Higher Task 2 score.

6. Soccer Goalie Deflection Bot

Arena:

- A 45 × 14 cm goal on a green carpet.
- A 10 cm baseline zone for the goalie.
- A penalty spot 1.2 m out.
- A ramp that shoots balls to the left, centre, or right.

Task performance (60 points):

- **Task 1 – Shot saves (20 points):**
  - 10 shots in a fixed mix of 3 left, 3 right, and 4 centre, released in random order. Each save scores 2 points.
  - Mistakes: –2 if the mechanism jams.

- **Task 2 – Angle tracking (20 points):**
  - 12 points for positioning correctly against 30°, 45°, and straight shots.
  - 8 points for not reacting to hand movements or background noise.
  - Mistakes: –3 for moving before the ball is released, and –5 for missing a ball entirely.
- **Task 3 – Reaction and return (20 points):**
  - 12 points for a reaction time of 150 ms or less.
  - 8 points for returning to centre within 1 second.
  - Mistakes: –4 for getting stuck in a corner.

Category rules:

- Each attempt is one series of 10 shots, fired 10 seconds apart.
- The goalie must stay within the baseline.

Category tie-breaker: More corner saves.

7. Rescue "Victim Sorter" Rover

Arena:

- A black line track on white, with 90° turns.
- Three 10 × 10 cm colour tiles:
  - **Red (harmed victim):** The rover drops a package.
  - **Green (safe victim):** The rover signals with an LED or buzzer only.
  - **Black (hazard):** The rover turns 180° and detours.
- A 15 cm drop circle next to the red tile.

Task performance (60 points):

- **Task 1 – Colour sorting (20 points):**
  - 7 points for red, 7 for green, and 6 for black.
  - Mistakes: –5 for a misread colour, and –3 for a false trigger on white.
- **Task 2 – Track traversal (20 points):**
  - 12 points for completing the track.
  - 8 points for centring over the tiles.
  - Mistakes: –3 per derailment, and –5 for a hand reset.
- **Task 3 – Package drop (20 points):**
  - 12 points for releasing the package.
  - 8 points for landing it fully inside the ring.
  - Mistakes: –4 if it rolls out, and –10 if it fails to drop or drops on the wrong tile.

Category rules:

- Each attempt can last up to 4 minutes.
- Teams get 60 seconds before each attempt to calibrate the colour sensor under the arena lighting.

Category tie-breaker: Higher Task 1 score.

8. Rescue "Victim Payload Gripper"

Arena:

- A 50 × 50 cm pick-up zone holding a ball (about 10 g) and a block (about 45 g).
- A 1 m approach path.
- A 15 × 15 cm platform, 6 cm high, with a 10 cm landing ring.

Task performance (60 points):

- **Task 1 – Detect and clamp (20 points):**
  - 8 points for detecting the victim automatically within 3 cm.
  - 8 points for a clean clamp.

  - 4 points for a soft grip on the ball.
  - Mistakes: –5 for knocking the object out, and –3 for repeated empty grabs.
- **Task 2 – Lift and transport (20 points):**
  - 10 points for lifting above 6 cm.
  - 10 points for carrying it 1 m without a drop.
  - Mistakes: –5 for a drop, and –3 for the arm sagging.
- **Task 3 – Platform deposit (20 points):**
  - 10 points for reaching over without a collision.
  - 10 points for landing inside the ring.
  - Mistakes: –4 if it rolls off, and –5 for freeing a jam by hand.

Category rules:

- Each attempt can last up to 4 minutes.
- Teams may use the counterweights supplied in the kit to stop the chassis tipping.

Category tie-breaker: Higher Task 1 score.

9. RF Swarm / Defender Bots

Arena:

- A 180 × 120 cm field.
- A master bot at midfield commands a slave bot defending 30 cm in front of goal.
- A moving ball for tracking tests.

Task performance (60 points):

- **Task 1 – Pairing, range, and failsafe (20 points):**
  - 6 points for linking within 2 seconds.
  - 6 points for no signal drop across the field.
  - 4 points for 20 clean packets.
  - 4 points for the slave halting if the link is lost for over 1 second.
  - Mistakes: –5 for a manual re-pair.
- **Task 2 – Swarm manoeuvres (20 points):**
  - 10 points for mirroring the master within 200 ms.
  - 10 points for carrying out the Defense, Attack, and Halt commands.
  - Mistakes: –3 for a lag over 500 ms, and –5 for moving in the wrong direction.
- **Task 3 – Defence wall (20 points):**
  - 12 points for forming a double block.
  - 8 points for blocking a shot.
  - Mistakes: –4 if the bots collide.

Category rules:

- Each attempt can last up to 4 minutes, including 60 seconds for pairing.
- Teams must be able to switch the RF channel if a nearby arena causes interference.

Category tie-breaker: Higher Task 2 score.

Notes for organisers

- **Section 10 is still empty.** The Rescue Maze "Wall-Hugger" category has no arena, tasks, or rules yet.
- **Code assumption.** This version assumes the code comes with the kit and teams only adjust calibration values. If teams are expected to write code, assembly time bands and the hour will need more room.
- **Removed criteria.** Budget checks, cost tie-breakers, custom app or web dashboard design, and the viva were removed, since kits and code are supplied and judging now focuses on assembly and mistakes. A short viva could be added back as bonus points if you want to test understanding.

- **Assembly time bands.** The same bands apply to every category, but the swarm (two bots) and ESP32 builds take longer. Consider adding 10 minutes to their bands.
- **Measuring time.** Reaction times of 150–200 ms are hard to judge by eye; a slow-motion phone video works well as a check.